

Introduction to Data Shaping and Grouping

INTRODUCTION TO POWER QUERY IN EXCEL

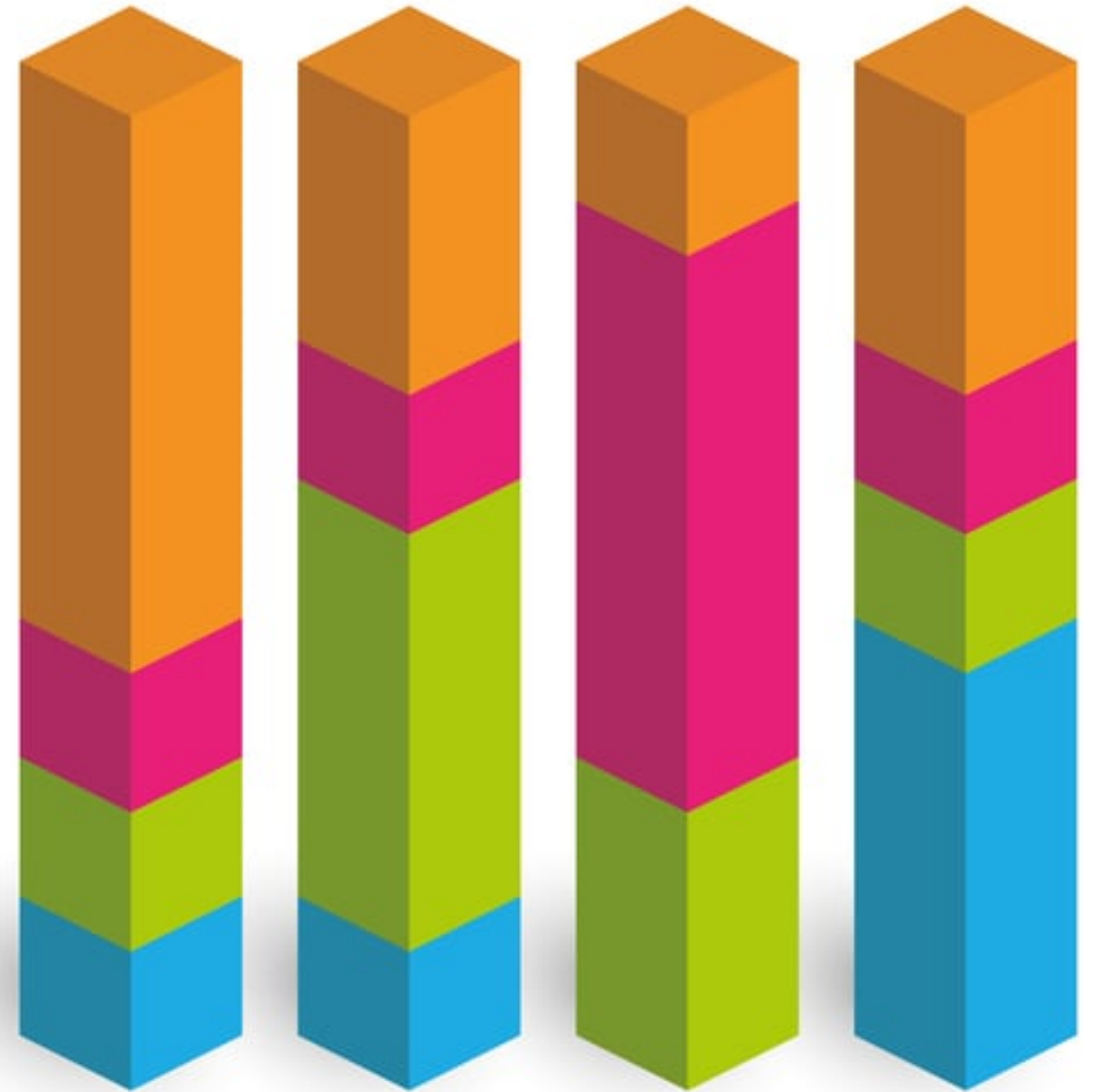


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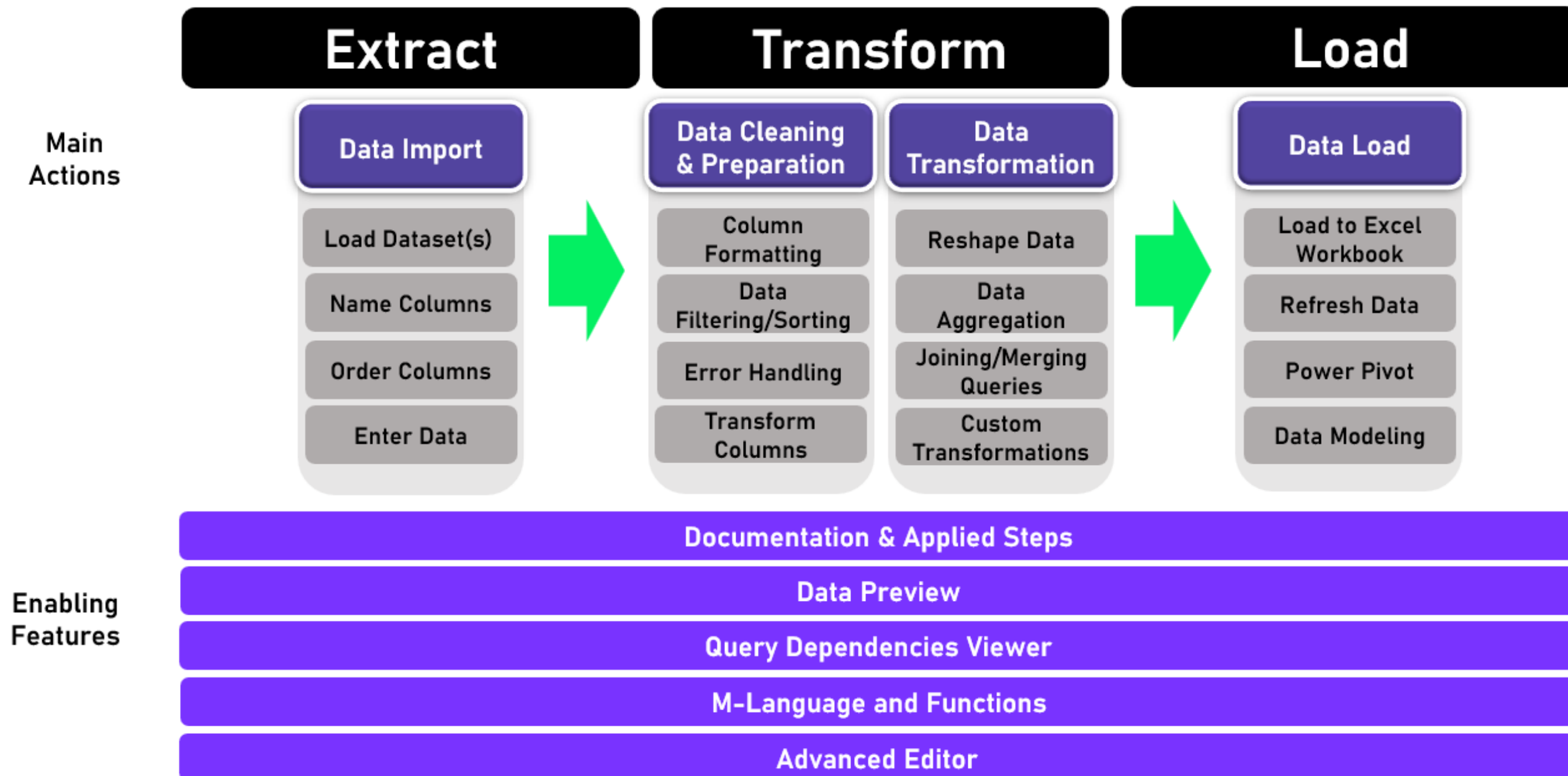
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Shaping and Grouping Data

- **Shaping:** restructuring data to make it more suitable for analysis.
- **Grouping:** Aggregating and summarizing data based on specific columns.



Excel Power Query



Excel Power Query

Main
Actions

Extract

Data Import

Load Dataset(s)

Name Columns

Order Columns

Enter Data

Transform

Data Cleaning
& Preparation

Column
Formatting

Data
Filtering/Sorting

Error Handling

Transform
Columns

Data
Transformation

Reshape Data

Data
Aggregation

Joining/Merging
Queries

Custom
Transformations

Load

Data Load

Load to Excel
Workbook

Refresh Data

Power Pivot

Data Modeling

Enabling
Features

Documentation & Applied Steps

Data Preview

Query Dependencies Viewer

M-Language and Functions

Advanced Editor

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Transpose

Reshape rows to columns and columns to rows.

Item	Section	Patient1	Patient2
1	A	0	10
2	A	10	0
3	A	10	1
4	B	0	1
5	B	1	0
6	B	0	10

Patient	Item1	Item2	Item3	Item4	Item5	Item6
Section	A	A	A	B	B	B
Patient1	0	10	10	0	1	0
Patient2	10	0	1	1	0	10

Group By

- Aggregate data into groups based on specific column values (e.g., sum).

Group By

- Aggregate data into groups based on specific column values (e.g., sum).
- Retains row and column orientation.

Item	Section	Patient1	Patient2
1	A	0	10
2	A	10	0
3	A	10	1
4	B	0	1
5	B	1	0
6	B	0	10

Section	Patient1	Patient2
A	20	11
B	1	11

Pivot

- Reshape data and create a cross-tabulation by rotating rows into columns.

Item	Section	Patient1	Patient2
1	A	0	10
2	A	10	0
3	A	10	1
4	B	0	1
5	B	1	0
6	B	0	10

Patient	Section A	Section B
Patient 1	0	0
Patient 1	10	1
Patient 1	10	0
Patient 2	10	1
Patient 2	0	0
Patient 2	1	10

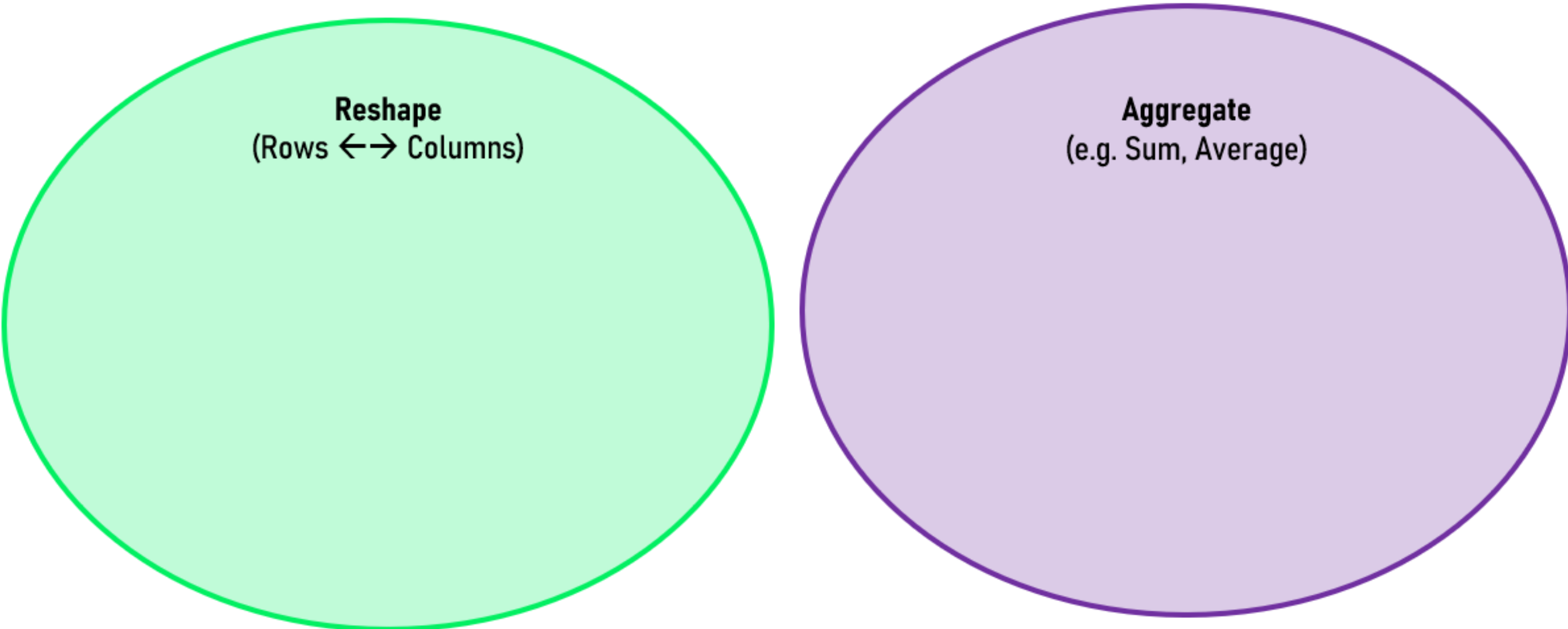
Pivot

- Reshape data and create a cross-tabulation by rotating rows into columns.
- AND group/aggregate data based on specific columns (e.g., sum).

Patient	Section A	Section B
Patient 1	0	0
Patient 1	10	1
Patient 1	10	0
Patient 2	10	1
Patient 2	0	0
Patient 2	1	10

Patient	Section A	Section B
Patient 1	20	1
Patient 2	11	11

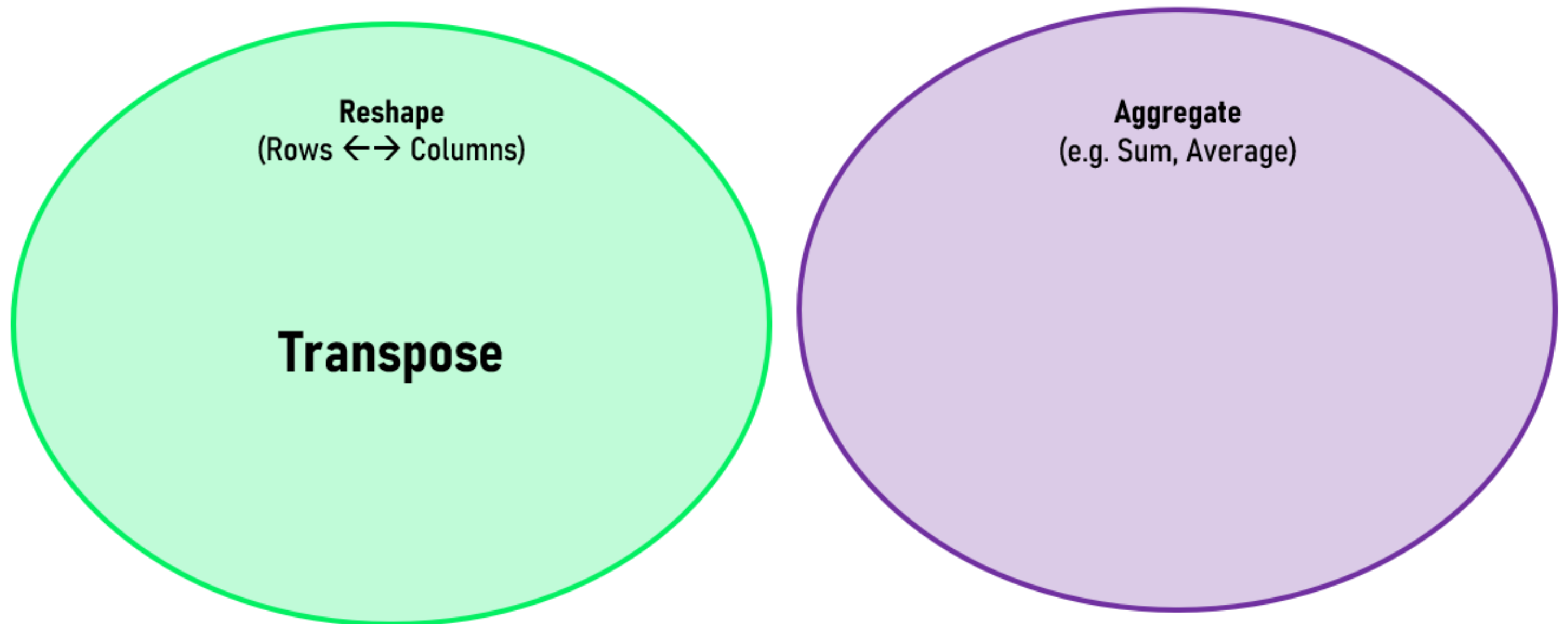
Transpose vs. Pivot vs. Group By



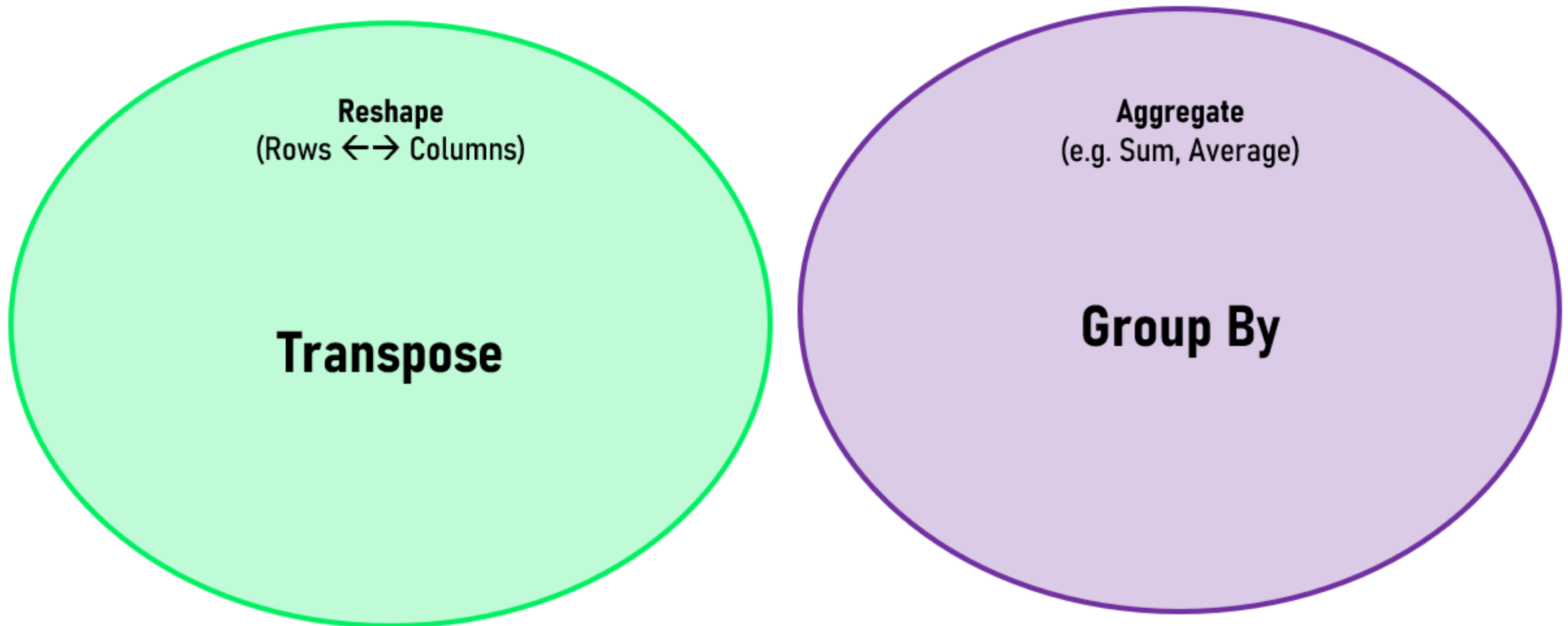
Reshape
(Rows $\leftrightarrow$ Columns)

Aggregate
(e.g. Sum, Average)

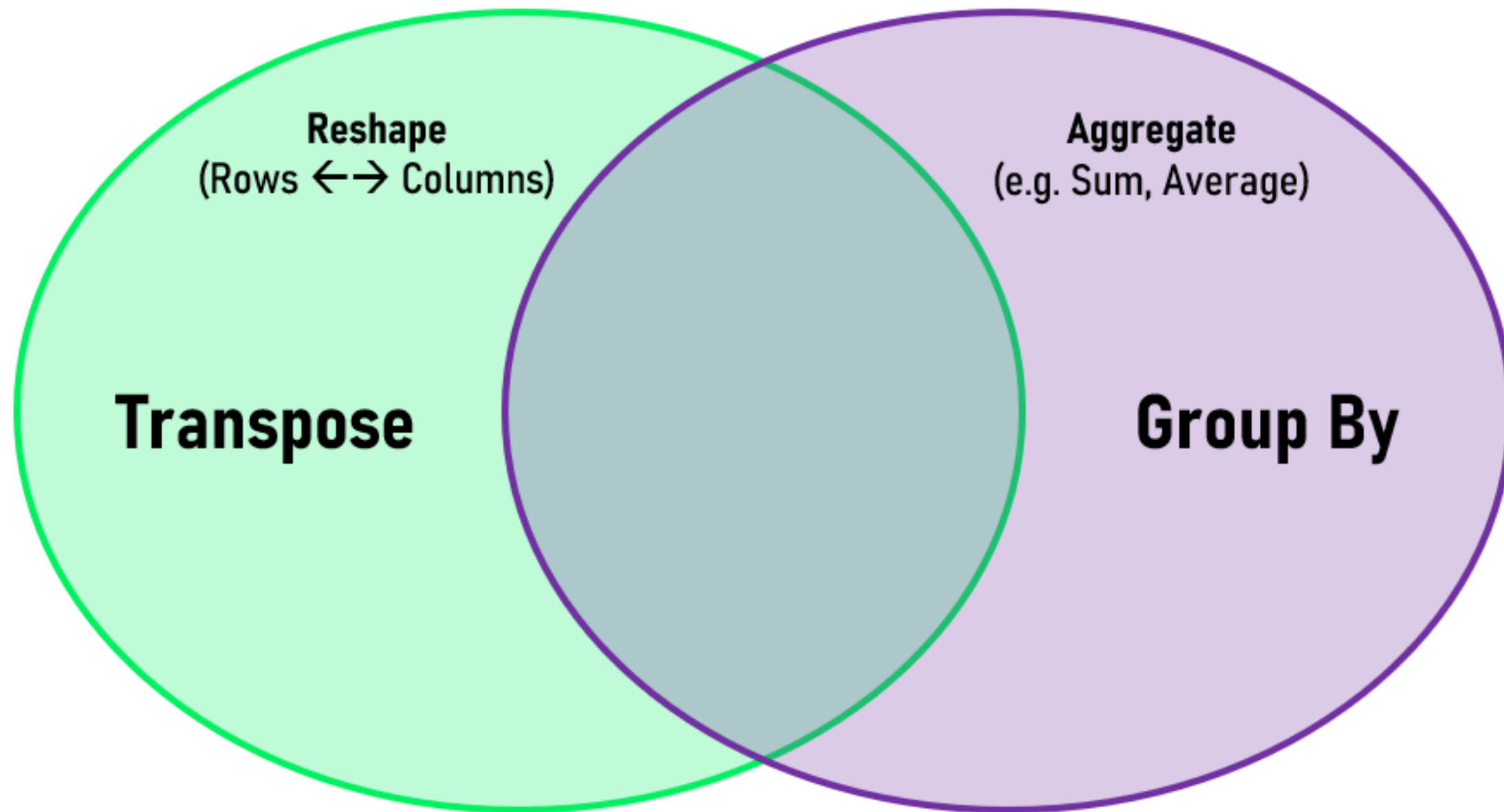
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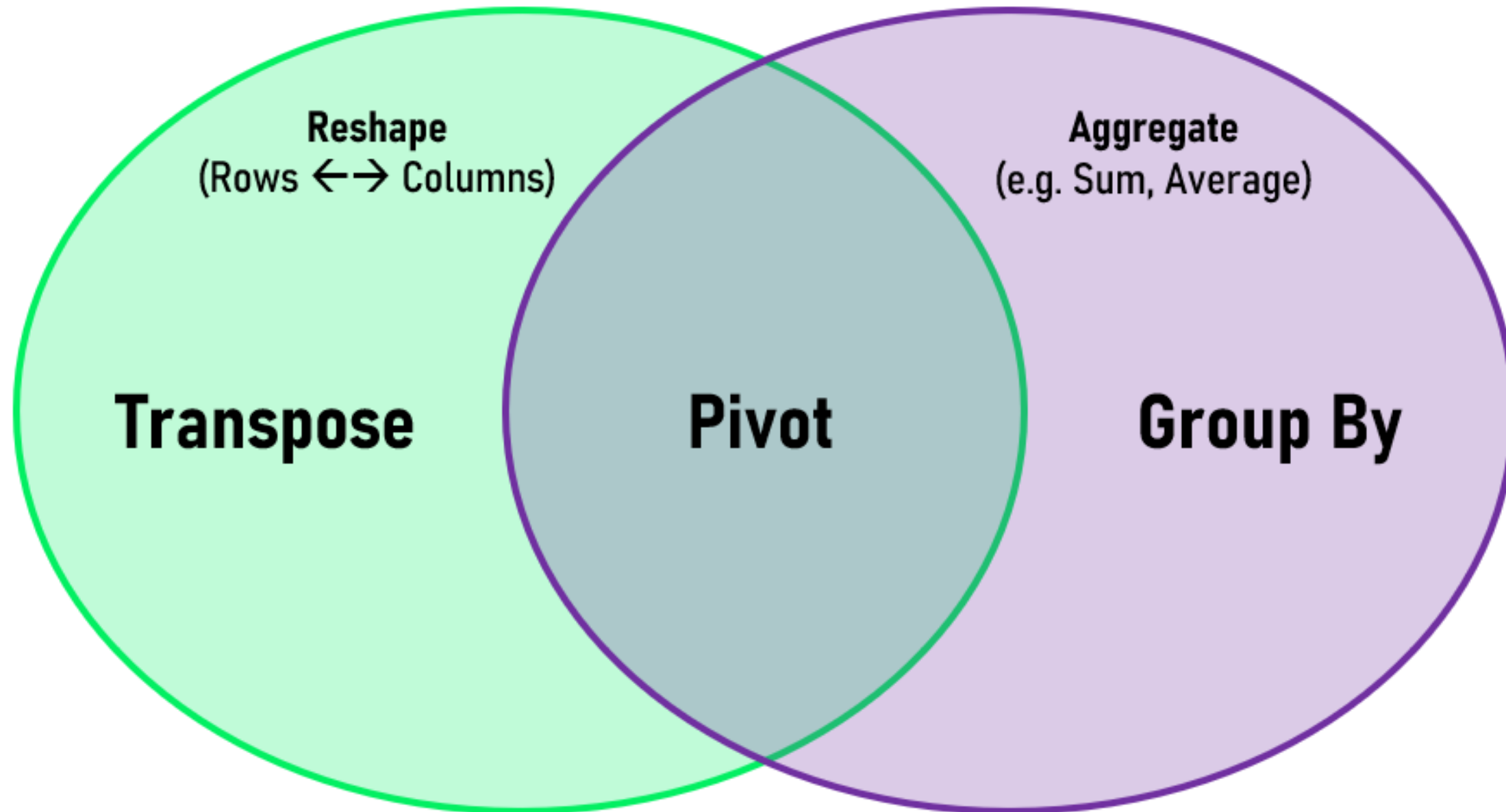
Transpose vs. Pivot vs. Group By



Transpose vs. Pivot vs. Group By



Transpose vs. Pivot vs. Group By



Let's practice!

INTRODUCTION TO POWER QUERY IN EXCEL

Shaping and Grouping Data in Excel Power Query

INTRODUCTION TO POWER QUERY IN EXCEL



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Let's practice!

INTRODUCTION TO POWER QUERY IN EXCEL

Congratulations!

INTRODUCTION TO POWER QUERY IN EXCEL

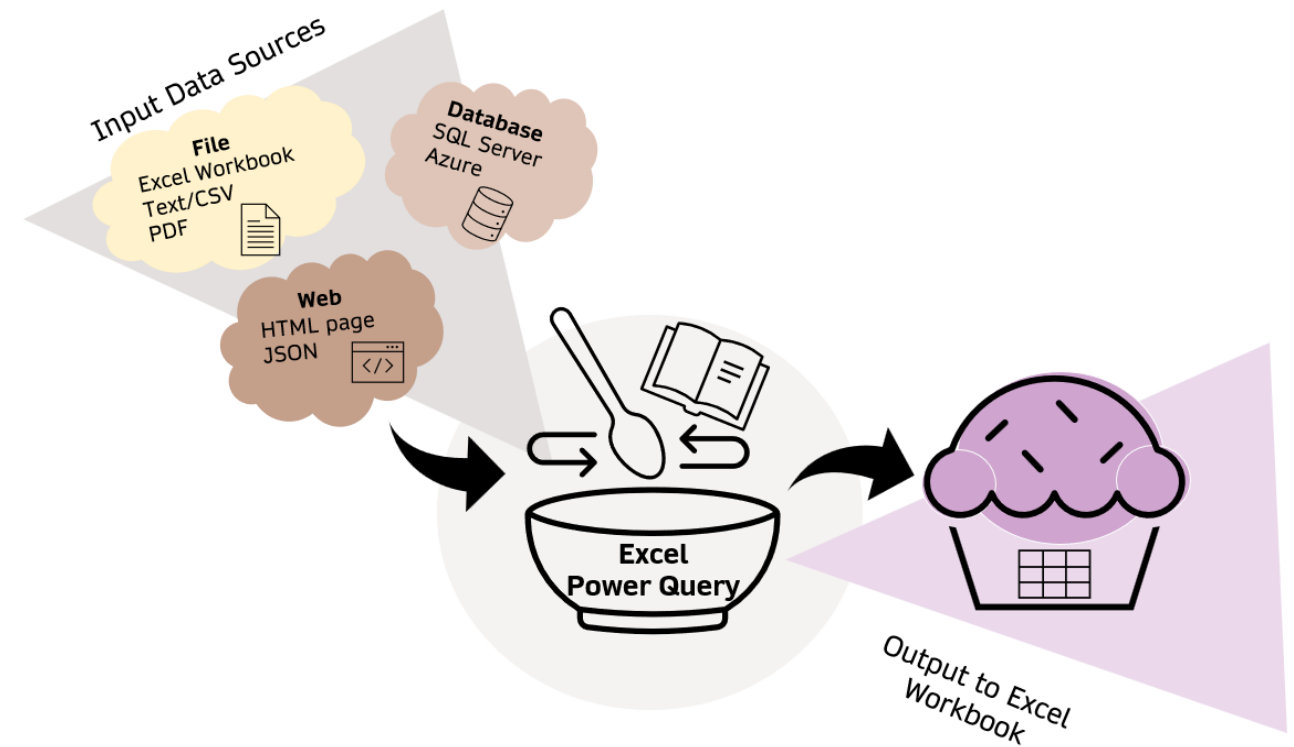


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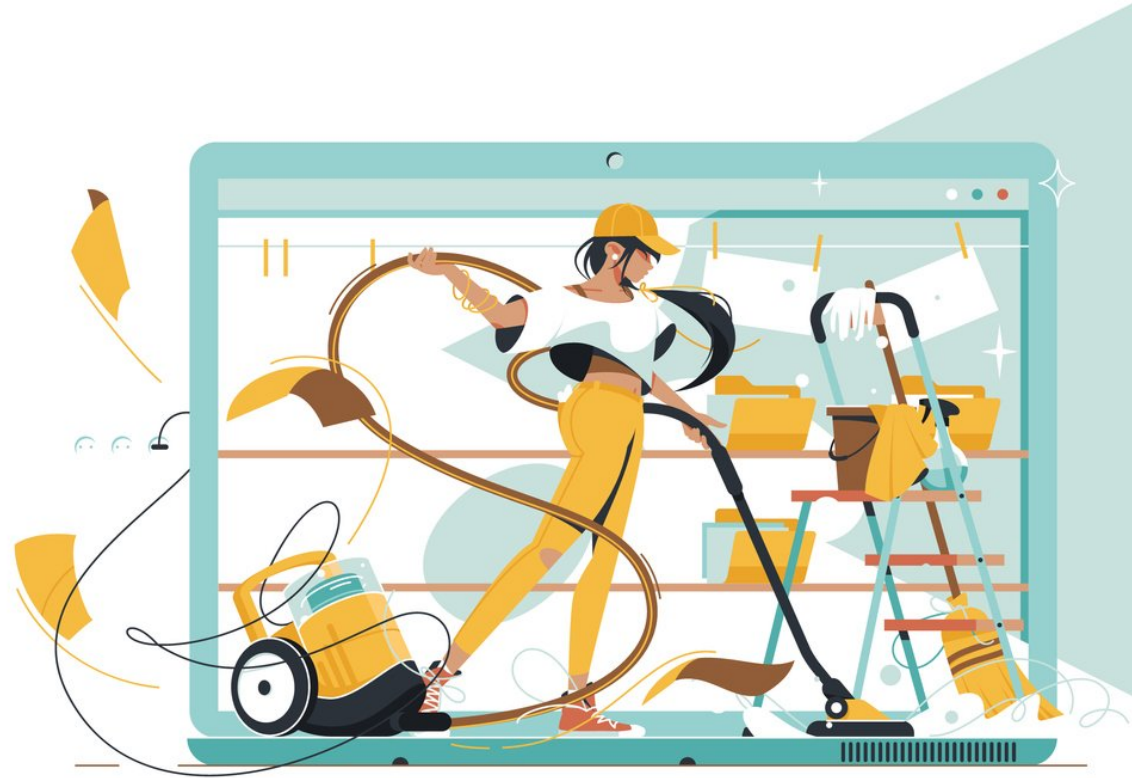
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Getting started with Power Query in Excel

- Main features of Power Query Editor.
- Power Query's role in ETL data workflows.
- Load data from multiple sources.
- Row and Column Manipulations.
- Applied Steps log.
- Loading tables into Excel Workbook.



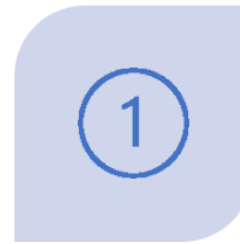
Data Cleaning and Preparation



- Column profile/view column features.
- Data filtering.
- Troubleshoot errors and missing data.
- Removing duplicates.
- Handling outliers.

Column Transformations

- Numerical transformations.
- Date/Time transformations.
- Text transformations.



NUMERIC

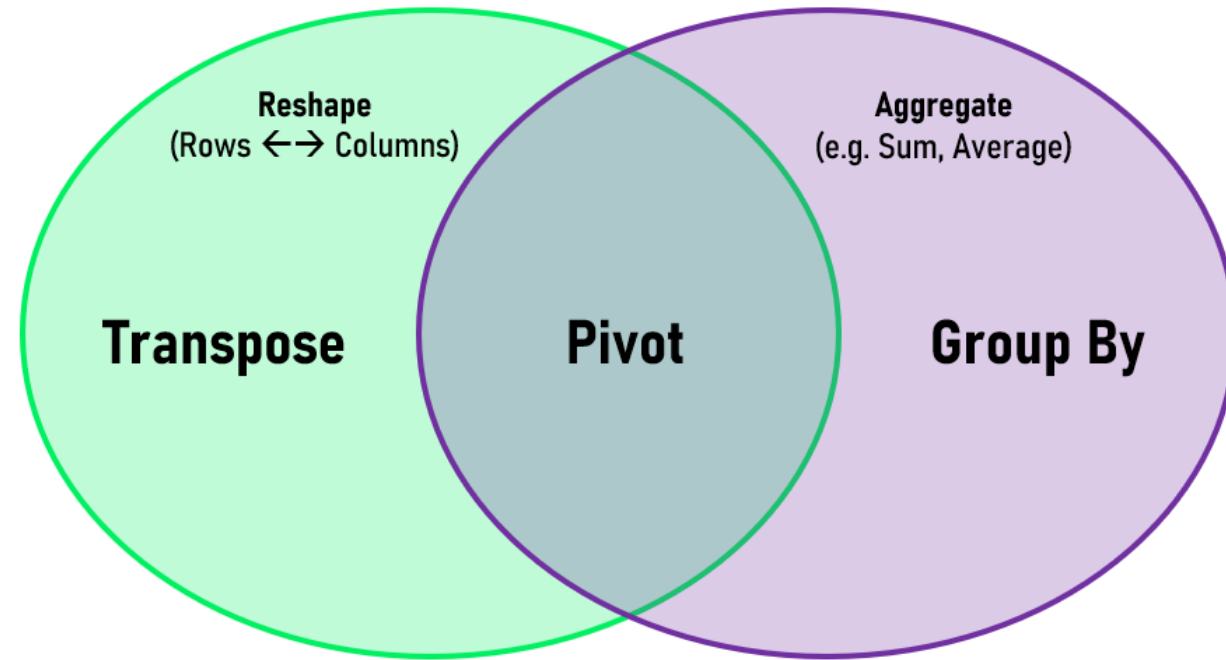


DATE/TIME



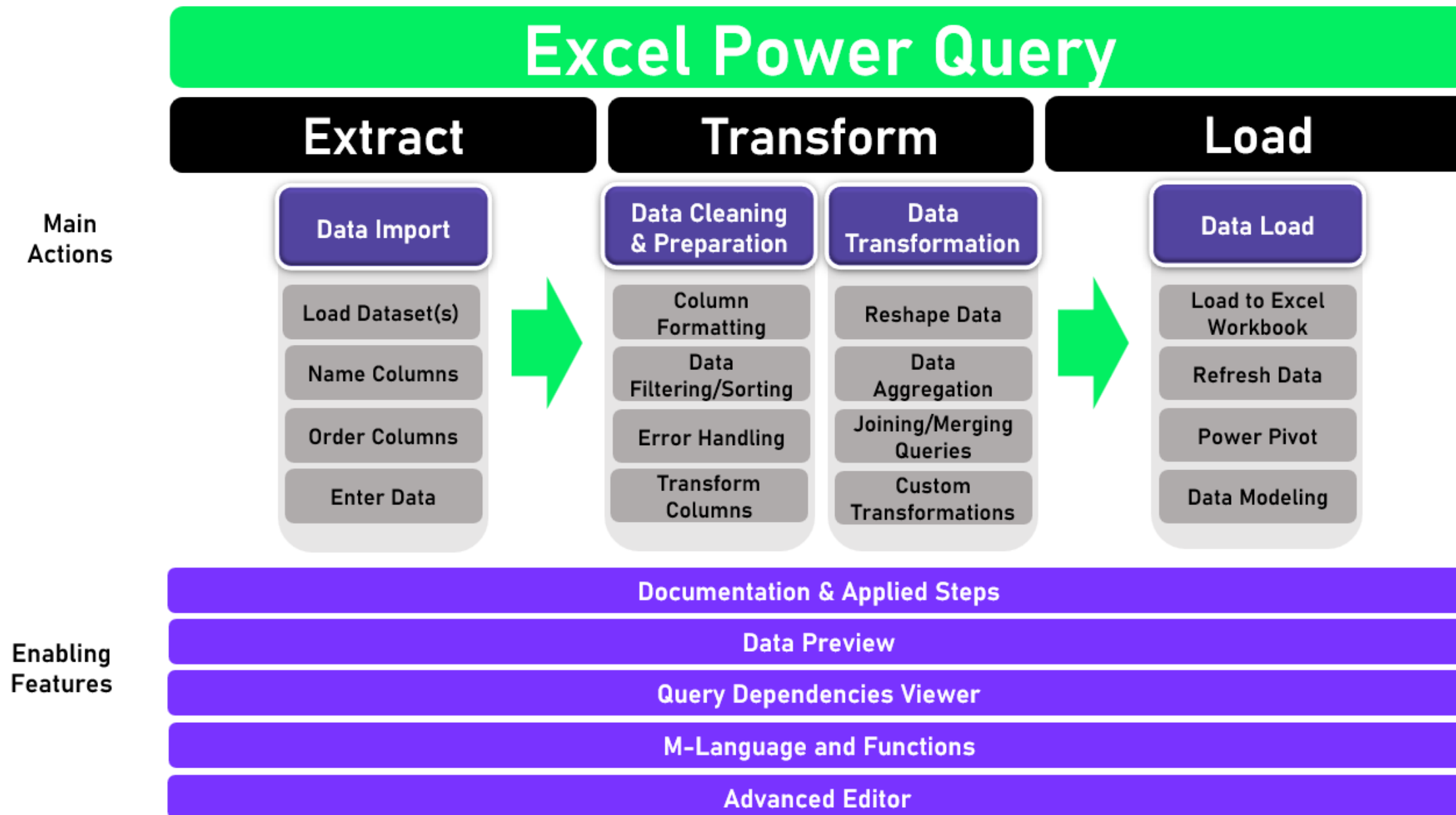
TEXT

Shaping and Grouping Data

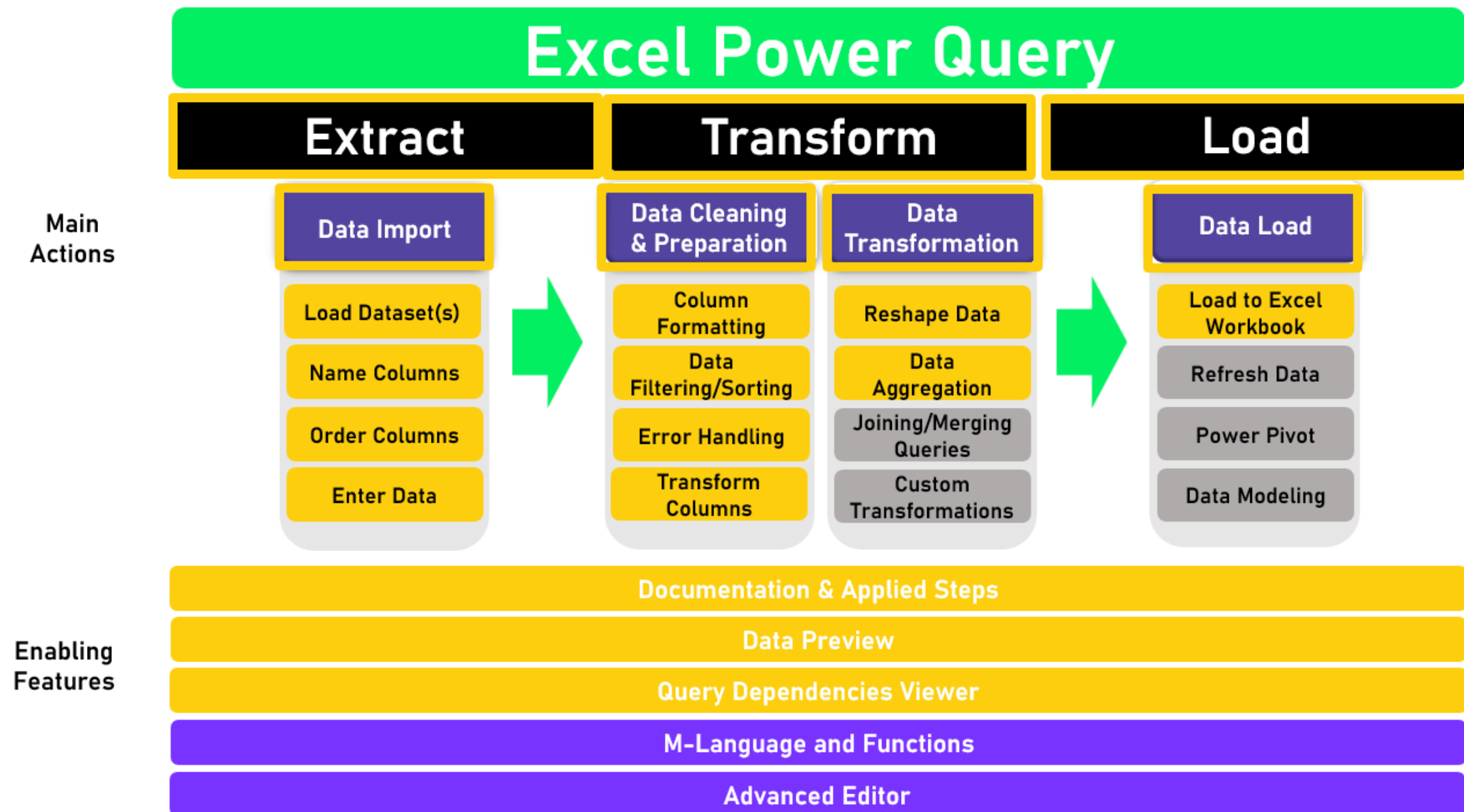


- Transpose transformations.
- Pivot transformations.
- Group By transformations.
- Reference vs. Duplicate query.

Where we started



What we have covered



Congratulations!

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